

1.

(1) 因为 X_i 是 X 的简单随机样本, $X_1, \dots, X_n$ 相互独立, $P(X_i = k) = \frac{\lambda^k e^{-\lambda}}{k!}$

$$\Rightarrow F(X_1, X_2, \dots, X_n) = F(X_1) \cdot F(X_2) \cdot \dots \cdot F(X_n) = \frac{\lambda^{\sum_{i=1}^n X_i}}{\prod_{i=1}^n X_i!} e^{-n\lambda}$$

$$(2) E(\bar{X}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = E(X) = \lambda$$

$$D(\bar{X}) = D\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n D(X_i) + \frac{1}{n^2} \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j) = \frac{\lambda}{n}$$

$$E(S^2) = D(X) = \lambda$$

$$E\left(\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right) = E\left(\frac{n}{n-1} S^2\right) = \frac{n}{n-1} \lambda$$

2. 排序 $(2, 2, 3, 3, 3, 3, 4, 5, 7, 9)^T$

$$x < 2, F_{10}(x) = 0$$

$$2 \leq x < 3, F_{10}(x) = 0.2$$

$$3 \leq x < 4, F_{10}(x) = 0.6$$

$$4 \leq x < 5, F_{10}(x) = 0.7$$

$$5 \leq x < 7, F_{10}(x) = 0.8$$

$$7 \leq x < 9, F_{10}(x) = 0.9$$

$$9 \leq x, F_{10}(x) = 1$$

$$\rightarrow \begin{cases} 0, & x < 2 \\ 0.2, & 2 \leq x < 3 \\ 0.6, & 3 \leq x < 4 \\ 0.7, & 4 \leq x < 5 \\ 0.8, & 5 \leq x < 7 \\ 0.9, & 7 \leq x < 9 \\ 1, & 9 \leq x \end{cases}$$

3. (1) $F(x) = \int_{-\infty}^x p(x) dx$, 由于 $p(x)$ 是偶函数, $\int_0^a p(x) dx = \int_{-a}^0 p(x) dx$, $\int_{-\infty}^0 p(x) dx = 0.5$

$$\int_{+\infty}^a p(x) dx = \int_{-a}^{-\infty} p(x) dx$$

$$\begin{aligned} F(-a) &= \int_{-\infty}^{-a} p(x) dx = \int_{-\infty}^0 p(x) dx + \int_0^{-a} p(x) dx = 0.5 - \int_0^a p(x) dx \\ &= \int_{-\infty}^{+\infty} p(x) dx + \int_{+\infty}^{-a} p(x) dx = 1 - F(a) \end{aligned}$$

$$(2) P(|X| < a) = F(a) - F(-a) = 2F(a) - 1$$

$$(3) P(|X| > a) = F(-a) + 1 - F(a) = 2 - 2F(a)$$

$$(4) \bar{X} \sim N(100, \frac{4}{10}), \bar{Y} \sim N(100, \frac{4}{15}), \bar{X} - \bar{Y} \sim N(0, \frac{7}{15})$$

$$P(|\bar{X} - \bar{Y}| > 0.2) = P\left(\frac{|\bar{X} - \bar{Y}|}{\sqrt{\frac{7}{15}}} > \frac{0.2}{\sqrt{\frac{7}{15}}}\right) = P(|Z| > 0.2928) \approx 2 \times 0.3859 = 0.7718$$

4. $X_1 + X_2 \sim N(0, 2\sigma^2)$, $X_1 - X_2 \sim N(0, 2\sigma^2)$

$$Y_1 = \frac{(X_1 + X_2)^2}{2\sigma^2} \sim \chi^2(1), \quad Y_2 = \frac{(X_1 - X_2)^2}{2\sigma^2} \sim \chi^2(1)$$

验证 Y_1, Y_2 相互独立

$$\Rightarrow \text{Cov}(Y_1, Y_2) = E(Y_1 Y_2) - E(Y_1) \cdot E(Y_2)$$

$$= E((X_1 + X_2)(X_1 - X_2)) = EX_1^2 - EX_2^2 = 0$$

得证, 则 $\left(\frac{X_1 + X_2}{X_1 - X_2}\right)^2 \sim F(1, 1)$

$$5. \bar{X}_n \sim N(\mu, \frac{\sigma^2}{n}), \frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2(n-1)$$

$$X_{n+1} - \bar{X}_n \sim N(0, \frac{\sigma^2}{n} + \sigma^2)$$

$\Rightarrow$ 若要使 $c \cdot \frac{X_{n+1} - \bar{X}_n}{S_n}$ 服从 t 分布

$$\text{则} \frac{X_{n+1} - \bar{X}_n}{\sqrt{\frac{n+1}{n}} \sigma} \times \frac{\sqrt{n-1} \cdot \sigma}{\sqrt{n-1} S_n} = \frac{X_{n+1} - \bar{X}_n}{S_n} \times \sqrt{\frac{n}{n-1}} \sim t(n-1)$$

$$\Rightarrow c = \sqrt{\frac{n}{n+1}}$$

b. 矩估计: $\frac{1}{n}(X_1 + X_2 + \dots + X_n) = \bar{X} = \frac{1}{p}$

$$\Rightarrow \hat{p} = \frac{1}{\bar{X}}$$

最大似然估计: $L(p) = p^n \cdot (1-p)^{\sum_{i=1}^n (X_i - 1)}$

$$\ln L(p) = n \ln p + \sum_{i=1}^n (X_i - 1) \ln(1-p) = f(p)$$

对其求导可得, $f'(p) = \frac{n}{p} + \frac{\sum_{i=1}^n X_i - n}{1-p}$

$$f'(p) = 0 \Rightarrow \hat{p} = \frac{n}{\sum_{i=1}^n X_i}$$

7.

矩估计: $E(X) = \int_0^1 x \cdot (\theta+1)x^\theta dx = \frac{\theta+1}{\theta+2} x^{\theta+2} \Big|_0^1 = \frac{\theta+1}{\theta+2}$

$$\bar{x} = \frac{\hat{\theta}+1}{\hat{\theta}+2} \Rightarrow \hat{\theta} = \frac{1-2\bar{x}}{\bar{x}-1}$$

代入得, $\hat{\theta} \doteq 0.3$

最大似然估计: $L(\theta) = (\theta+1)^n \left(\prod_{i=1}^n x_i \right)^\theta$, $\ln L(\theta) = n \ln(\theta+1) + \theta \sum_{i=1}^n \ln x_i = f(\theta)$

求导得 $f'(\theta) = \frac{n}{\theta+1} + \sum_{i=1}^n \ln x_i$

令其 $f'(\theta) = 0 \Rightarrow \hat{\theta} = -1 - \frac{n}{\sum_{i=1}^n \ln x_i}$

代入得, $\hat{\theta} \doteq 0.2$

8. 首先判断是否具有无偏性, 即 $E(\hat{\sigma}^2) = \sigma^2$

$$E(\hat{\sigma}_1^2) = \frac{1}{n} \cdot n \cdot E(X^2) = E(X^2) = (EX)^2 - DX = \sigma^2$$

$$\begin{aligned} E(\hat{\sigma}_2^2) &= \frac{1}{n-1} \cdot n \cdot E(X - \bar{X})^2 = \frac{n}{n-1} \cdot [E(X - \bar{X})^2 - D(X - \bar{X})] \\ &= \frac{n}{n-1} \cdot (1 - \frac{1}{n}) \sigma^2 = \sigma^2 \end{aligned}$$

再验证有效估计

$$D(\hat{\sigma}_1^2) = \frac{1}{n^2} \sum_{i=1}^n D(X_i^2) = \frac{1}{n^2} \cdot n \cdot 2\sigma_1^4 = \frac{2\sigma_1^4}{n}$$

, 等于 Cramér-Rao 下界, 有效估计

$$D(\hat{\sigma}_2^2) = \frac{2\sigma^4}{n-1}, \text{ 大于下界, 不是有效估计}$$

9.

$$(1) \text{ 即验证 } E(\sum_{i=1}^n a_i X_i) = E(X)$$

$$\text{由于 } X_i \text{ 之间彼此独立, 则 } E(\sum_{i=1}^n a_i X_i) = \sum_{i=1}^n a_i E(X_i) = \sum_{i=1}^n a_i E(X) = E(X)$$

得证, $\sum_{i=1}^n a_i X_i$ 是 $E(X)$ 的无偏估计

$$(2) D(\sum_{i=1}^n a_i X_i) = D(X) \cdot \sum_{i=1}^n a_i^2$$

在 $\sum_{i=1}^n a_i = 1$ 的约束下, 由柯西-施瓦茨不等式可得

$$\Rightarrow \sum_{i=1}^n a_i^2 \geq \frac{1}{n}, \text{ 当且仅当 } a_i = \frac{1}{n} \text{ 时成立}$$

即 $a_i = \frac{1}{n}$ 时, 方差最小. 此时 $\sum_{i=1}^n a_i X_i = \bar{X}$, 得证

10.

由题意得, S^2 已知, 求 μ 置信度为 95% 的置信区间,

样本方差服从 $n-1=15$ 的卡方分布, $\alpha=0.05$

查表得, $t_{0.025}(15) = 2.131$

置信区间公式为: $\bar{x} \pm t_{0.025}(15) \cdot \frac{S}{\sqrt{n}}$

代入得 $(2.68955, 2.72045)$, 四舍五入得 $(2.69, 2.72)$

11.

(1) μ 未知, σ^2 未知, 拒绝域为 $(-\infty, \mu_0 - \frac{s}{\sqrt{n}} t_{\alpha}(n-1))$

$$\Rightarrow \mu_0 - \frac{s}{\sqrt{n}} t_{\alpha}(n-1) = 0.5 - \frac{0.037}{\sqrt{10}} t_{0.05}(9) = 0.4786$$

因为 $0.452 < 0.4786$, 落入拒绝域, 拒绝 H_0 , 即 $\mu < 0.5\%$

(2) 转化为 $H_0': \sigma = 0.04\%$; $H_1': \sigma < 0.04\%$

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} \xrightarrow{H_0'} \frac{(n-1)s^2}{(0.04\%)^2} \sim \chi^2(9)$$

由 $\alpha = 0.05$, $n = 10$ 得拒绝域为 $W': \chi^2 < \chi_{1-\alpha}^2(n-1) = \chi_{0.95}^2(9) = 3.325$

$$\text{又 } s = 0.037\%, \text{ 得 } \chi^2 = \frac{(10-1) \times (0.037\%)^2}{(0.04\%)^2} = 7.7006 \in W'$$

$\Rightarrow$ 接受 H_0'

12. 设 X_i 为入学初分数, Y_i 为训练后分数, 差值 $d_i = Y_i - X_i$

$$\bar{d} \doteq 4.33, S_d \doteq 7.94$$

$$t \text{ 统计量为 } t = \frac{\bar{d}}{S_d/\sqrt{n}} \doteq 1.64, \text{ 自由度 } df = 8$$

临界值 $t_{0.05}(8) \doteq 1.86$, 因 $t < t_{0.05}(8)$,

所以没有足够证据表明训练后有显著进步

13. 设零假设 $H_0: p \leq 0.02$, 备择假设 $H_1: p > 0.02$

样本大小 $n = 480$, 废品数 $X = 12$, 样本废品率 $\hat{p} = \frac{12}{480} = 0.025$

检验统计量 $Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$, 其中 $p_0 = 0.02$, 计算得 $Z \approx 0.784$

显著性水平 $\alpha = 0.05$, 临界值 $Z_\alpha = 1.645$

由于 $Z < Z_\alpha$, 不拒绝 H_0

$\Rightarrow$ 这批产品可以被接受

14.

由题意得, 总样本量 $n=109$, $n_1=10$, $n_2=53$, $n_3=46$

$$\hat{p} = \frac{2n_1 + n_2}{2n} \approx 0.33486$$

$$E_1 = n \cdot p^2 \approx 12.222$$

$$E_2 = n \cdot 2p \cdot (1-p) \approx 48.475$$

$$E_3 = n \cdot (1-p)^2 \approx 48.223$$

$$\text{卡方检验: } \chi^2 = \sum_{i=1}^3 \frac{(O_i - E_i)^2}{E_i} \approx 0.404 + 0.422 + 0.102 = 0.928$$

$$\text{自由度为1, 临界值 } \chi_{0.05}^2(1) = 3.841 > \chi^2$$

无法拒绝零假设, 因此数据与模型相符

15. 样本均值 $\bar{x} = 4.3$, 样本方差 $S^2 = 94.26$, 标准差 $S \approx 9.71$
预计其遵循分布 $N(4.3, 9.71^2)$

合并小频数组, 计算 $\chi^2 \approx 5.74$, 自由度 $k-1-r = 4$

查表得 $\chi_{0.05}^2(4) = 9.488$

$\Rightarrow \chi^2 < 9.488$, 接受原假设, 认为尺寸偏差服从正态分布